

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			LHS: $^{235}_{92}\text{U}$ (1) RHS: $^{140}_{55}\text{Cs} + ^{92}_{37}\text{Rb} + 4\ ^1_0\text{n}$ Cs data correct (1) Rb data correct (1) FOUR neutrons (1)		4		4	2	
	(b)	(i)		To slow <u>neutrons</u> down (1) so {cause fission of / are absorbed by / are captured by / react with} <u>uranium</u> (1)	2			2		
		(ii)		By control rods (1) being raised (1) accept removed so fewer neutrons are absorbed [by control rods] or more neutrons available to be absorbed by <u>uranium</u> (1)	1	1 1		3		
	(c)	(i)		65 536 [g] (1) No rounding accepted		1		1	1	
		(ii)	I	16		1		1	1	
			II	8 half-lives remaining (1) Sequence of halving starting at 512 [g] (1) Award the 1 st and 2 nd marks for: $\frac{1}{2^8}$ = 2 [g] (1) Award 2 marks for an answer of 4 [g]		3		3	3	

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		(iii)		Rubidium decays faster / rubidium has a shorter half-life / less rubidium than caesium remaining as time passes (1) Accept the converse argument So ratio increases / after 64 seconds the ratio becomes 32 768:1 (1) Accept attempt of numerical calculation that shows an increase in the ratio			2	2	1	
				Question 6 total	3	11	2	16	8	0